MediTrust SRS Addendum: Patient Safety Layer

Project: MediTrust **Version:** 1.1 (Addendum to SRS v1.0, Buildathon MVP) **Document type:** Software Requirements Specification addendum, structured after IEEE 830 / ISO 29148

This addendum extends SRS v1.0. Everything in v1.0 still applies unless this document says otherwise. Requirement IDs here use new prefixes so they never clash with v1.0.

1. Introduction

1.1 Purpose

SRS v1.0 describes a general track-and-trace and clone-detection system. Its skeleton (batch Merkle root, custody chain, one-time sale record, public verification) is generic and could be reused for coffee, wine, or electronics. This addendum defines the features that make MediTrust specifically a **patient-safety platform for medicines in Ethiopia**, so that it is clearly different from supply-chain provenance projects.

1.2 Differentiation statement

> Provenance systems tell you where a product came from. MediTrust protects the patient: it makes recalls actually reach people, lets anyone verify a medicine in Amharic from the app they already use, and gives pharmacies a verifiable reason to be trustworthy.

Coffee and similar goods have no regulator-driven recall, no life-and-death stakes, no pharmacist gatekeeper, and no expiry-based danger. These features exist only because medicine has them.

1.3 New pillars

Pillar	One-line description
P1. Recall Reach	Recalls notify the patients who scanned the pack, and pharmacies must prove they quarantined stock

Pillar	One-line description
P2. Verify Anywhere	Telegram bot and SMS-style flow with Amharic text and voice read-out
P3. Pharmacy Trust Score	Public, on-chain-derived score for each pharmacy
P4. Counterfeit Early-Warning Map	Live map of scan anomalies, unknown spikes and reports
P5. National Interoperability Adapter	GS1-style import adapter (mocked in MVP)

1.4 Scope

In scope: the five pillars above, the contract and database changes they require, and updated demo and acceptance criteria.

Out of scope: real SMS gateway contracts with telecoms, payments, real EFDA integration, any storage of patient identity on-chain.

1.5 Priority key

M = must have, S = should have, C = could have. Because time is limited, Section 9 recommends a minimum set.

2. Pillar 1: Recall Reach and Acknowledgement

2.1 Problem

In v1.0 a recall only changes a flag on-chain. A patient who already owns the pack, or a pharmacy that does not log in, may never find out. A recall that does not reach people is not a safety feature.

2.2 Functional requirements

ID	Requirement	Priority
FR-RR-01	After scanning a pack, the verification page offers an optional "Alert me if this medicine is recalled" action	M

ID	Requirement	Priority
FR-RR-02	The patient chooses a channel (Telegram, SMS number, or email) and gives explicit consent; no account or wallet is needed	M
FR-RR-03	The system stores only the contact (encrypted at rest), the batch ID, and a consent timestamp; the serial is not linked to the contact	M
FR-RR-04	When a `BatchRecalled` event is detected, the system sends each subscriber of that batch a short alert in their chosen language with the reason and what to do	M
FR-RR-05	A patient can unsubscribe at any time with one tap, and all their stored contact data is deleted	M
FR-RR-06	Every distributor and pharmacy holding stock of a recalled batch must call `acknowledgeRecall(batchId, quantityQuarantined, evidenceCid)`	M
FR-RR-07	The acknowledgement records quantity quarantined, a timestamp, and an optional photo or document hash on IPFS	M
FR-RR-08	The regulator console shows a **recall compliance view**: for each holder, status (Acknowledged, Pending, Overdue), quantity held versus quantity quarantined, and time since recall	M
FR-RR-09	A recall has a configurable acknowledgement deadline (default 48 hours); holders past the deadline are marked Overdue on-chain by anyone calling `markOverdue`	S
FR-RR-10	Holders who have not acknowledged receive repeated dashboard banners, and optionally a Telegram or SMS reminder	S
FR-RR-11	A recall can carry a severity level (Class I, II, III) that controls alert wording and deadline	C

2.3 Smart contract additions

```
struct RecallAck { uint32 quantityQuarantined; uint64 timestamp; string
evidenceCid; }
mapping(uint256 batchId => mapping(address holder => RecallAck)) public
recallAcks;
mapping(uint256 batchId => uint64) public recallDeadline;

function acknowledgeRecall(uint256 batchId, uint32 qty, string evidenceCid)
// Distributor, Pharmacy
function markOverdue(uint256 batchId, address holder)
// anyone, after deadline
event RecallAcknowledged(uint256 indexed batchId, address indexed holder,
uint32 qty, uint64 time);
event RecallOverdue(uint256 indexed batchId, address indexed holder, uint64
time);
```

Rules:

1. Only an approved holder with a non-zero holding of a recalled batch can acknowledge.
2. An acknowledgement cannot exceed the holder's recorded holding.
3. An acknowledgement cannot be edited, only added to by a later, larger entry.
4. `markOverdue` reverts if the holder already acknowledged or the deadline has not passed.

2.4 Off-chain components

A small worker (Next.js route plus scheduled job, or a Supabase edge function) listens for `BatchRecalled` and `RecallLifted` events and drives notifications. It must not be required for the recall to be effective on the verification page.

3. Pillar 2: Verify Anywhere (Telegram and Amharic Voice)

3.1 Problem

Many users in Ethiopia have limited data, older phones, and low literacy in English. A PWA that needs a camera and a good connection excludes them. Telegram is widely used, works on low bandwidth, and is quick to integrate.

3.2 Functional requirements

ID	Requirement	Priority
FR-VA-01	A Telegram bot lets a user send a photo of a pack's QR code or paste the code text	M
FR-VA-02	The bot decodes the QR, calls the same `verifyPack` read function as the web app, and replies with the result, product name, batch number and expiry date	M
FR-VA-03	The reply uses the standard status colors as emoji (green, blue, red, orange, dark red) plus a plain sentence; never color alone	M
FR-VA-04	The bot works in English and Amharic; the user picks the language with a command or button and the choice is remembered	M
FR-VA-05	The bot offers a **voice reply**: an Amharic audio message reading the result aloud	S
FR-VA-06	The bot offers buttons: "Report a problem", "Alert me about recalls", "Find a verified pharmacy"	S
FR-VA-07	If the QR cannot be decoded, the bot guides the user to retake the photo or type the code	M
FR-VA-08	The bot logs each check like a public scan (FR-VER-05): no Telegram identity or phone number is stored with the scan record	M
FR-VA-09	The verification logic is shared in one library used by both the PWA and the bot so results can never disagree	M
FR-VA-10	A printable **SMS-format spec** is documented (for example `VERIFY <batch>.<serial>.<code>`) as the design for a future SMS or USSD gateway, with a mock endpoint in the MVP	C

3.3 Voice read-out

Pre-recorded Amharic audio clips for the five status results are acceptable for the MVP, with product names read as text. Text-to-speech for Amharic may be used if a reliable service is available; verify quality with a native speaker before the demo.

3.4 Constraints

- The bot has no wallet and no keys; it only reads from the chain.
- Proof length must fit the chosen QR format; the bot uses the same payload as the web app.
- Message length stays short (under about 400 characters) for readability.

4. Pillar 3: Pharmacy Trust Score

4.1 Problem

Pharmacies have little reason to join a verification system if joining only creates extra work. Patients have no way to choose a pharmacy that handles medicines safely.

4.2 Functional requirements

ID	Requirement	Priority
FR-TS-01	Each approved pharmacy has a public profile showing a Trust Score from 0 to 100 and a badge level (Bronze, Silver, Gold)	M
FR-TS-02	The score is computed only from verifiable data: on-chain events and regulator actions, never patient reviews alone	M
FR-TS-03	The score formula and each pharmacy's breakdown are visible to everyone	M
FR-TS-04	Score changes are explainable: the profile lists the events that raised or lowered the score	S
FR-TS-05	Patients can search or filter pharmacies by badge level and see them on a map	S

ID	Requirement	Priority
FR-TS-06	The regulator can see a ranked list and a list of pharmacies below a configurable threshold	M
FR-TS-07	A pharmacy cannot delete or hide its score, but can see how to improve it	M

4.3 Scoring model (MVP)

Start at 50. Recompute off-chain from chain events; the formula is deterministic and published.

Signal	Effect
Recall acknowledged within deadline	+ up to 20
Recall overdue	up to -30
Clone attempt caught at its own counter (`DuplicateSaleAttempt` by this pharmacy, correctly blocked)	+ 2 each, capped at +10
Custody-versus-sales gap: packs held, past a time window, never sold or transferred, and then found sold elsewhere	up to -20
Confirmed counterfeit report naming this pharmacy	-25 each
Dismissed report naming this pharmacy	no change
Months active with no incident	+ 1 per month, capped at +10
Expiry hygiene: stock sold or quarantined before expiry	up to +10

Badge levels: Gold 80 or more, Silver 60 to 79, Bronze below 60. The exact weights are tunable configuration, not hard-coded.

4.4 Fairness and abuse

- A single anonymous report must never lower a score; only regulator-confirmed reports count.

- Scores are labeled "Based on recorded activity in MediTrust" so they are not mistaken for a government rating.
- The pharmacy may add a response note to any negative event (stored as an IPFS hash).

5. Pillar 4: Counterfeit Early-Warning Map

5.1 Functional requirements

ID	Requirement	Priority
FR-EW-01	The regulator console shows a live map layer of recent scans with result status, aggregated to about 1 km cells	M
FR-EW-02	A heatmap layer shows concentration of Unknown results and confirmed or pending reports	M
FR-EW-03	Clicking an area lists the flags, batches and reports involved	S
FR-EW-04	Anomaly rule R4 (Unknown spike) creates an alert card with batch number, count, and area	M
FR-EW-05	A new rule R5 flags a **cluster**: three or more Unknown results from different devices within 5 km in 24 hours	S
FR-EW-06	A new rule R6 flags a **sold-elsewhere pattern**: a serial reported Sold by one pharmacy but scanned Genuine-unsold at another location	S
FR-EW-07	The regulator can turn an alert into an investigation and optionally trigger a recall in one click (extends FR-RPT-04)	S
FR-EW-08	A read-only public "Safety Pulse" page shows aggregate counts only, with no pack-level detail	C

5.2 Privacy

Locations are rounded as in v1.0. Device hashes are salted and non-reversible. The public page never shows individual scans.

6. Pillar 5: National Interoperability Adapter

6.1 Purpose

To answer "does this already exist?" with a working demonstration that MediTrust can plug into a national system rather than compete with it.

6.2 Functional requirements

ID	Requirement	Priority
FR-IO-01	A mock **national registry** service exposes a small API returning product records keyed by a GS1-style identifier (GTIN, batch, expiry)	S
FR-IO-02	An adapter module cross-checks a MediTrust batch against the mock registry when a manufacturer registers a batch	S
FR-IO-03	The verification page can show "Also registered in the national registry" when the check matches	S
FR-IO-04	If the registry is unreachable, verification still works (the check is additive, never blocking)	M
FR-IO-05	The adapter is a documented interface so a real EFDA or GS1 feed could replace the mock	C

The mock must be clearly labeled as simulated in the interface and in the demo. Do not claim real integration.

7. Updated data model

7.1 New and changed off-chain tables

Table	Key fields
recall_subscriptions	id, batch_id, channel (telegram/sms/email), contact_encrypted, language, consent_at, active
notification_log	id, subscription_id, batch_id, sent_at, status (sent/failed)
recall_compliance_cache	batch_id, holder, status, qty_held, qty_quarantined, acked_at
pharmacy_scores	pharmacy, score, badge, breakdown_json, computed_at
score_events	id, pharmacy, event_type, delta, ref_tx, created_at
bot_sessions	telegram_id_hash, language, created_at
scan_events (changed)	add: channel (web/telegram/mock_sms)
geo_cells	cell_id, lat_center, lng_center, unknown_count, genuine_count, window_start
registry_mock	gtin, product_name, batch_number, expiry, source

7.2 Privacy rules for new data

1. Contact details are stored only with consent, encrypted, and deleted on unsubscribe.
2. A subscription links to a **batch**, never to a serial or a specific scan.
3. Telegram IDs are stored only as salted hashes for session language; the notification table holds the delivery handle only while the subscription is active.
4. Nothing in this addendum puts personal data on-chain.

8. Non-functional requirements (additions)

ID	Category	Requirement
NFR-21	Reliability	A recall is effective on-chain and on the verification page even if the notification worker is down
NFR-22	Performance	Recall alerts are queued within 60 seconds of the `BatchRecalled` event in the demo
NFR-23	Performance	Bot reply within 5 seconds of receiving a decodable QR image
NFR-24	Consistency	Bot and web verification use one shared library; a test confirms identical results for the same inputs
NFR-25	Privacy	Unsubscribing deletes contact data within 24 hours and shows confirmation
NFR-26	Explainability	Every Trust Score change links to its cause (event or regulator action)
NFR-27	Accessibility	Voice replies and text replies carry the same information
NFR-28	Localisation	All bot messages, alerts and score labels live in translation files (extends NFR-10)
NFR-29	Safety	Notification text is short, non-alarmist, states the batch and the action to take, and tells the user to contact a pharmacist or health facility, never to self-treat

9. Prioritization and recommended minimum

If time is short, do not attempt everything. Build in this order and stop when the time runs out:

1. **Recall acknowledgement (FR-RR-06 to FR-RR-08).** This is purely contract and dashboard work, uses existing roles, and directly separates MediTrust from provenance projects.

2. **Telegram bot with Amharic text (FR-VA-01 to FR-VA-04, FR-VA-09).** High demo impact and low cost.
3. **Patient recall alerts through Telegram (FR-RR-01 to FR-RR-05).** Pairs naturally with the bot; the same message channel is reused.
4. **Pharmacy Trust Score (FR-TS-01 to FR-TS-03, FR-TS-06).** A simple version needs only a script over chain events.
5. **Early-warning map (FR-EW-01, FR-EW-02, FR-EW-04).** Extends the existing regulator map.
6. **Amharic voice, interoperability adapter and everything marked C.**

Signature combination for the pitch: Recall Reach + Telegram verification in Amharic + Pharmacy Trust Score.

10. Updated demo scenario (3 minutes)

Replaces steps 6 to 9 of the v1.0 demo and keeps steps 1 to 5.

1. Steps 1 to 5 as in v1.0: problem, approvals, register and ship, genuine scan, sale.
2. **Telegram scan (20 s):** a patient sends a photo of the QR to the bot and receives a green result in Amharic, with an optional voice reply. They tap "Alert me about recalls".
3. **Clone attack (25 s):** a photocopied QR is presented at a second pharmacy; the red clone alert appears and the attempt is logged on-chain. The early-warning map lights up.
4. **Fake QR (10 s):** an invented code shows Unknown; two more from nearby create a cluster alert (R5) on the regulator map.
5. **Recall (30 s):** the regulator recalls the batch. The patient's phone receives the Telegram alert within seconds. The regulator's compliance view shows two holders Pending.
6. **Acknowledgement and score (25 s):** one pharmacy acknowledges and quarantines stock; the other goes Overdue. The Trust Scores update, and the patient sees one Gold and one Bronze pharmacy.
7. **Close (10 s):** "Provenance tells you where it came from. MediTrust makes sure a recall reaches the patient, and the evidence can't be erased."

11. Updated acceptance criteria

In addition to the v1.0 criteria, the addendum is accepted when the following can be demonstrated:

- A recalled batch shows every holder as Pending in the regulator compliance view until each calls `acknowledgeRecall`
- A holder with no stock of the batch cannot acknowledge, and an acknowledgement cannot exceed holdings (tested)
- A patient who subscribed through the verification page or bot receives a recall alert, and unsubscribing stops alerts and deletes the contact
- The Telegram bot returns the same result as the web app for the same QR (genuine, sold, recalled, expired, unknown), in English and Amharic
- The bot works without the database for the verification result itself (chain RPC only)
- A pharmacy profile shows a Trust Score with a visible breakdown, and a recall overdue event lowers it
- One anonymous report does not change any score; a regulator-confirmed report does
- The regulator map shows a heatmap and an alert card when more than 10 Unknown results hit one batch in 24 hours
- If the mock national registry is offline, verification still completes
- All new contract functions have unit tests, including: acknowledging without holdings, acknowledging a non-recalled batch, double acknowledgement, overdue before deadline, overdue after acknowledgement

12. Risks and mitigations (additions)

Risk	Mitigation
Scope too large for the buildathon	Follow the Section 9 order; treat anything marked C as cuttable
Telegram is not available to everyone	Position it as the first channel; document the SMS and USSD design as the next step
Patients distrust giving a phone number	Offer Telegram and email options; never link contact to a serial; clear consent text; one-tap delete

Risk	Mitigation
Unfair pharmacy scores or false accusations	Only regulator-confirmed events count; formula public; right of reply note
Amharic voice quality is poor	Use pre-recorded clips reviewed by a native speaker; keep text as the primary output
Judges say the national system already covers this	Interoperability adapter plus the patient-facing recall reach, which a national hub does not currently offer to patients
Legal and privacy rules on messaging and health data	Minimal data, consent, testnet and mock data only; verify applicable rules before any real pilot

13. Build plan additions

Phase	Output
1b Contracts	`acknowledgeRecall`, `markOverdue`, recall deadline, events, tests
3b Shared verify library	One package used by the PWA and the bot
8b Telegram bot	Webhook route, QR decode, multilingual replies
8c Recall alerts	Event listener, subscriptions table, notification sender
7b Compliance view	Regulator recall compliance table
8d Trust Score	Score script, pharmacy profile badge, regulator ranking
8e Early-warning map	Geo-cell aggregation, heatmap, alert cards
9b Interop mock	Mock registry and adapter
9c Demo	Seed data covering the updated scenario

14. Traceability to v1.0

This addendum	Extends v1.0
FR-RR	FR-REC-01 to FR-REC-05, FR-DSH-03, FR-DSH-04
FR-VA	FR-VER-01 to FR-VER-08, FR-LOC-01
FR-TS	FR-REG-05, FR-REG-06
FR-EW	FR-VER-06, FR-RPT-03, FR-RPT-04, Rules R1 to R4
FR-IO	Section 11 future integration point